

Introduction to ESL Modelling

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SoC Architecture Development and Modelling

SoC Design Flow

- Frequent use of high-level C or C++ models
- Prototyping a new IP block (TPU example)
- Convert C Model to RTL by hand

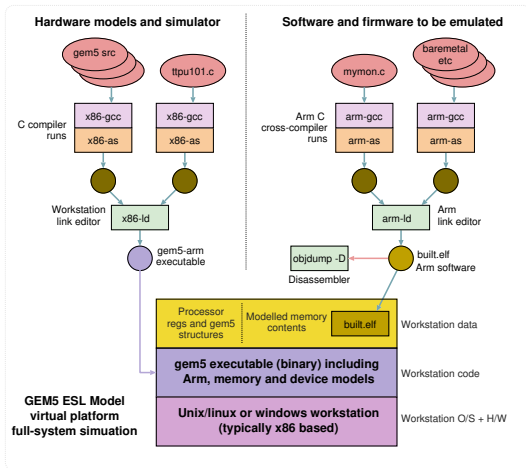
Other tools:

- Could alternatively use an HLS (high-level synthesis) tool for "C to gates".
- Verilator: converts RTL to a C model.

C or C++ Model Roles

- Fast logic simulator - cycle-accurate only and only using 0 and 1 (not X or Z)
- Prototyping a new IP block behaviour (TPU example)
- A component in a fast ESL model.

Electronic Systems Level (ESL) Model



A full-system model:

- Estimate overall system performance
- Explore bus traffic patterns
- Develop software and firmware

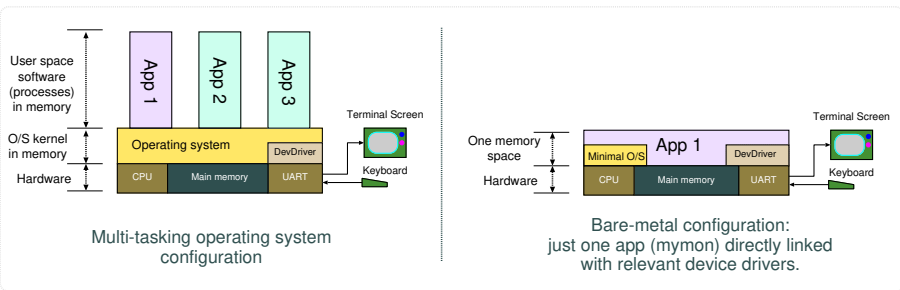
Adding a new IP block might require off-chip DRAM storage and read/write bandwidth.

Changing the hardware models and structure to assess performance is known as **Architectural Exploration**

Adding new IP block to ESL Model

Open source GEM5 provides various ESL models, including Arm.
In this workshop:

- We add our model to GEM5 Virtual Platform
- Fully implement PIO (programmed I/O) access
- Use a bare-metal machine code monitor to exercise it
- (The same software setup as used in the first Skills Workshop)



We have just one device driver, for the UART, but in a typical system there would be a further driver for the new IP block.

Architectural Exploration

Want to know:

- is the design is functionally correct?
- Primary: what is its data throughput (bytes per clock)?
- Secondary: how much silicon is used?
- Secondary: how much energy/power is used?

Architectural Exploration is the process of making large changes to the design structure to see what gets better and what gets worse.

ESL Model Aims

Primary aim to rapidly explore consequences of design changes:

- Adding a new IP block or another processor
- Adding a DMA controller
- Reorganising the bus structures
- Repartitioning the work between the various processors.

The ESL model tells us the correctness of each design.

Software Development Role

- An ESL model is a virtual platform or simulator on which the complete software stack can be run.
- Hence, software developers can make progress before the working silicon is delivered.

Modelling Style

Different styles: high accuracy or high performance.

An accurate model might be orders of magnitude slower than real time.

For instance, Skills Video 1 showed GEM5 was nearly a 100x slower than real time.

Coding Styles

- Pure C/C++ for algorithm/functionality correctness
- Free-standing (timing-annotated?) cycle-callable models
- Coded inside open-source platforms (QEMU or GEM5)
- SystemC - free C++ library that models hierarchy and synchronous logic.
- SystemC TLM - transactional level library for subroutine calling between components instead of net wiggling.
- Other, in-house proprietary tools.

Summary

Ideally: A connected route from high-level model through to tape out.

A good ESL modelling system enables an arbitrary mixture of

- high-level models (for speed)
- low-level implementations (for accuracy or checking)

to be combined.

All combinations give a full-system executable system, but modelling in detail only in the areas of current interest.